HHS Cyber

HTOC

HEALTHCARE THREAT OPERATIONS CENTER

Daily Partner Prediction Report Consolidation

Version 1.0
April 2026

Table of Contents

Table of Contents

1. Purpose

This SOP describes the daily consolidation of per-partner NOI (Next Observed Indicator) prediction CSVs into a single full daily report. Each day, partner-specific prediction files are written to subfolders under the `OpDiv_Predictions` directory. This process discovers those files, merges them into one consolidated dataset, and writes a dated output CSV to the `Full Daily Reports` folder.

This process runs as **Python** (`main.py` on a schedule, or the appendix standalone copy for ad hoc runs).

-

2. Scope

This procedure applies to HTOC analysts and data engineers who run, monitor, or troubleshoot the daily prediction report consolidation. It covers scheduled execution, manual script runs, and troubleshooting.

-

3. Prerequisites

3.1 Environment

Requirement	Notes
Python 3.x	Must be available in the environment where \`main.py\` executes
pandas	CSV loading and concatenation
numpy	Indirect dependency
Access to \`Z:\HTOC\Data_Analytics\\\`	All input and output paths are on this share

Example dependency install command:

```
pip install pandas
```

3.2 Input Data

Partner prediction CSVs must be present under the `OpDiv_Predictions` directory tree **before** this process runs. These files are produced by the NOI forecasting pipeline (scheduled job and/or forecasting script documented in `SOP_NextObservedIndicator_Forecasting.md`).

Expected input file naming convention:

```
Z:\HTOC\Data_Analytics\Data\OpDiv_Predictions\{PartnerName}\{YYYYMMDD}.csv
```

-

4. Key Configuration

Production `main.py` and the appendix standalone script use the same path constants:

Constant	Value
\`DATA_PATH\`	\`Z:\HTOC\Data_Analytics\Data\OpDiv_Predictions\`

Constant	Value
`SAVE_PATH`	`Z:\HTOC\Data_Analytics\Data\OpDiv_Predictions\Full Daily Reports`
`EXCLUDE_FOLDERS`	`automation scripts`, `Logs`, `LogsBackup`, `Full Daily Reports`

The `EXCLUDE_FOLDERS` set prevents the loader from reading its own output or log directories during the folder walk.

-

5. How the Process Works

5.1 File Discovery

The loader uses `os.walk` to recursively traverse all subfolders under `DATA_PATH`. For each directory, it:

Skips any path containing a folder name in `EXCLUDE_FOLDERS`.

Treats the **immediate folder name** as the `Partner` value.

Matches files whose names end with `{YYYYMMDD}.csv` — in the automated path, only **today's** date is matched.

5.2 Consolidation

For each matched file:

The CSV is read into a DataFrame.

Two columns are added: `Partner` (from the folder name) and `FileDate` (from the filename date string).

All per-partner DataFrames are concatenated into a single `daily_search` DataFrame.

If no files are found, the process prints `"No CSV files found for today."` and exits without writing an output file.

5.3 Output

The consolidated DataFrame is written to:

```
Z:\HTOC\Data_Analytics\Data\OpDiv_Predictions\Full Daily
Reports\full_daily_report_{YYYYMMDD}.csv
```

The output file is **never overwritten** — if it already exists for today's date, the process prints `"File already exists: …"` and skips the write. This prevents duplicate runs from corrupting the report.

-

6. Automated Execution (main.py)

6.1 Schedule

Production `**main.py**` is maintained on **SharePoint**: [NextObserved/main.py](https://hhsgov.sharepoint.com/:u:/r/sites/HTOCDataAnalyticsASA/Shared%20Documents/HTOC%20Data%20Analytics/Python%20Scripts/NextObserved/main.py?csf=1&web=1&e=qebIUV). When scheduled on **F.R.E.D**, Task Scheduler normally runs this script every day at **8:15 AM**. It executes the **today-only** consolidation flow: load today's partner CSVs → merge → save single consolidated report. The GitHub copy under `Next Obs Daily/src/` is a mirror unless your team intentionally runs from Git.

Manual execution (**Section 7**) is only necessary when troubleshooting a failed scheduled run, backfilling historical dates, or validating a revised `main.py` before redeploy.

6.2 Script Logic

```
main()
  ├── Compute today_str (YYYYMMDD)
  ├── load_all_csvs_from_folders(DATA_PATH, today_only=True)
  │       ├── Walk DATA_PATH, skip EXCLUDE_FOLDERS
  │       ├── Match files: {today_str}.csv only
  │       ├── Add Partner and FileDate columns
  │       └── Return concatenated DataFrame (or empty if none found)
  └── If DataFrame is non-empty:
          └── save_daily_report(df, SAVE_PATH, today_str)
                  ├── Build output path: full_daily_report_{today_str}.csv
                  ├── If file exists → print and skip
                  └── Else → makedirs + to_csv + print path
```

6.3 Verifying a Successful Automated Run

Confirm the run succeeded by checking that:

The file `full_daily_report_{YYYYMMDD}.csv` exists in `Full Daily Reports` for today's date.

The file is non-empty and contains the expected `Partner` and `FileDate` columns.

-

7. Manual execution (Python)

Use a **local copy** of production `main.py` (SharePoint) or the appendix **`daily_reports_standalone.py`** when you need to run consolidation outside Task Scheduler—for example after a failed schedule, spot checks, or while testing a code change before publish.

7.1 Verify prerequisites

[ ] `Z:\HTOC\Data Analytics\` is accessible and mapped.

[ ] Partner prediction CSVs for the target date exist under `Z:\HTOC\Data Analytics\Data\OpDiv Predictions\`.

[ ] Python 3.x with `pandas` (and any imports your deployed `main.py` requires).

7.2 Run today-only consolidation

From a shell, with the script on disk (path per your deployment):

```
py "path\to\main.py"
```

Expected behavior matches **Section 6.2**: non-empty partner files for today produce `full_daily_report_{YYYYMMDD}.csv` under `Full Daily Reports` unless that file already exists.

7.3 Backfill or custom date ranges

The scheduled `main.py` is **today-only**. For historical rebuilds, either (a) temporarily adjust the date filter in a **non-production copy** of the script and run once per needed date with care, or (b) extend a private copy of the loader logic to iterate `FileDate` values—always validate outputs before replacing production archives.

-

8. Output File

Detail	Value
Location	`Z:\HTOC\Data_Analytics\Data\OpDiv_Predictions\Full Daily Reports\`
Filename	`full_daily_report_{YYYYMMDD}.csv`
Overwrite behavior	Never overwritten — skipped if already exists
Key added columns	`Partner` (source folder name), `FileDate` (from filename)

The output file contains all prediction columns from the upstream NOI forecasting output, with `Partner` and `FileDate` appended for traceability.

-

9. Troubleshooting

Symptom	Likely Cause	Resolution
`"No CSV files found for today."`	Partner prediction CSVs not yet written for today	Verify the NOI forecasting pipeline ran successfully; check `Z:\HTOC\Data_Analytics\Data\OpDiv_Predictions\` for today's files
`"File already exists: …"`	Automated run already completed successfully	No action needed; this is the expected idempotency guard
`"No data to save."`	`daily_search` is empty after loading	Same as above — check upstream prediction files
Output file is missing partners	One or more partner subfolders had no matching file	Check the partner's subfolder for a `{YYYYMMDD}.csv` file; the loader skips unreadable files but prints a skip message
`Skipping {path}: …` printed	A CSV file could not be read (encoding, permissions, corruption)	Inspect the file at the printed path; re-run after fixing or removing the corrupt file
Custom backfill needed	Built-in script is today-only	Use a controlled copy of the script with explicit date logic; see **Section 7.3**
`Z:\` path not found	Network drive not mapped	Map the `Z:\` drive to `\\cscso1fsappv01\...` and re-run

-

10. Relationship to Other Processes

This process sits **downstream** of the NOI forecasting pipeline and **upstream** of any reporting or distribution that consumes the consolidated daily file.

Step	Process	Output
1	NOI forecasting pipeline runs first	Partner-level daily files: `Z:\HTOC\Data_Analytics\Data\OpDiv_Predictions\{PartnerName}\{YYYYMMDD}.csv`
2	Daily reports consolidation (`main.py`) runs second	Consolidated daily file: `Z:\HTOC\Data_Analytics\Data\OpDiv_Predictions\Full Daily Reports\full_daily_report_{YYYYMMDD}.csv`
3	Downstream reporting/distribution jobs consume the consolidated file	Partner distribution products and operational reports

If the daily report is missing or incomplete, check the NOI forecasting step first before rerunning this process.

-

11. Appendix - Standalone Python Script

The consolidated runnable copy for ad hoc use is maintained at:

`H:\HTOC\documentation\SOP\Appendix Scripts\daily_reports_standalone.py`

Run with:

```
&"C:\Program Files\Python313\python.exe" "H:\HTOC\documentation\SOP\Appendix
Scripts\daily_reports_standalone.py"
```

-

12. Related Documents

`SOP_NextObservedIndicator_Forecasting.md` — upstream process that generates the per-partner input CSVs

`Z:\HTOC\Data Analytics\Data\OpDiv Predictions\` (input data location)

`Z:\HTOC\Data Analytics\Data\OpDiv Predictions\Full Daily Reports\` (output location)

SharePoint ([NextObserved/main.py](https://hhsgov.sharepoint.com/:u:/r/sites/HTOCDataAnalyticsASA/Shared%20Documents/HTOC%20Data%20Analytics/Python%20Scripts/NextObserved/main.py?csf=1&web=1&e=qebIUV)) — authoritative production batch script · GitHub: [`Next_Obs_Daily/src/main.py`](https://github.com/jaytlinaskew-OIS/HTOC/blob/main/scripts/batch-processing-script/Next_Obs_Daily/src/main.py) (mirror)